

وكالة الإمارات للفضاء
UAE SPACE AGENCY

أكاديمية الفضاء الوطنية
NATIONAL SPACE ACADEMY

SPACE42



Introductory Webinar · Driving impact through innovation

Arab Youth Space Hackathon

2026 Edition: Challenge 813

A regional innovation program that turns hyperspectral and multi-source Earth Observation data into real solutions for agriculture, water, climate and cities across the Arab world.

📅 30-minute session

💬 15-minute live Q&A

- Agenda

What we'll cover in the next 30 minutes

01



Overview & background

What the hackathon is and why it exists

02



Who is eligible to apply

Profiles, team rules and youth quota

03



The journey & what's unique

From an idea to a deployable product

04



The challenge themes

Six problem areas, with examples

05



Data and tools

The imagery, sensors and platform you get

06



Criteria for the PoC

What a Proof of Concept must show

07



Registration process

How, and by when, to apply

Then: open Q&A

Then: open Q&A

15 minutes, drop questions in the chat any time

- Overview

A months-long innovation program, not a weekend hackathon

The Arab Youth Space Hackathon turns real satellite data into solutions for the region's most pressing challenges. Teams work with experts, use the Space42 GIQ platform, and build a Proof of Concept, and the strongest teams continue into incubation to grow it into a startup or a pilot deployment.



6

official challenge themes



6

EO data types: hyperspectral, VHR, multispectral, SAR, thermal, LiDAR



22

Arab nations invited to take part



7

months from first idea to a deployable MVP

✓ Fully online across three phases

✓ Mentoring at every step

✓ Backed by UAESA, the National Space Academy and Space42

- Background

Why the "813" challenge exists

The program is named for the **Arab Hyperspectral Satellite 813**, the first Arab-collaboration satellite of its kind. Its launch marks a milestone in regional Earth observation: advanced hyperspectral sensing gives unprecedented spectral detail for analysing land, water, atmosphere and ecosystems across the Arab world.

This hackathon celebrates that mission by putting real Earth Observation data into the hands of young talent, and turning it into scalable, policy-relevant solutions for the region.

The mission

A regional hyperspectral Earth-observation satellite serving environment and climate.

The invitation

Build on that capability, and show what the region's youth can do with it.



PLACEHOLDER: SATELLITE 813 RENDER

Official satellite or launch image, to be supplied

- Organisers & partners

Who runs the program

A partnership that brings together national vision, talent development and the GIQ geospatial platform.

وكالة الإمارات للفضاء UAE SPACE AGENCY

Official Sponsor

UAE Space Agency

The federal body driving the UAE's space sector, providing strategic sponsorship and funding through the National Space Fund and aligning the program with national priorities.

أكاديمية الفضاء الوطنية NATIONAL SPACE ACADEMY

Lead Organizer

National Space Academy

A federal initiative established in 2023 under the UAE Space Agency; the national platform for space talent, leading the program's governance and direction.

SPACE42

Strategic Execution Partner

Space42

A UAE-based AI-powered SpaceTech company that operates GIQ, delivering the platform, operations, mentoring and participant support.

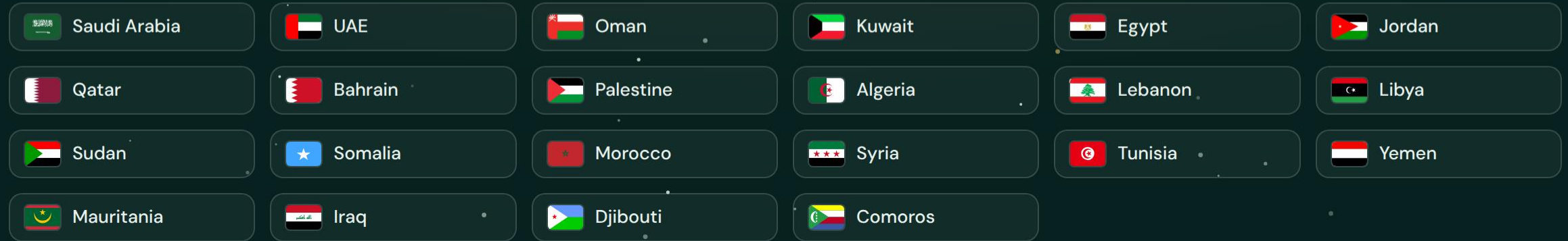
Supporting partners



- A pan-Arab community

Open to talent from all 22 Arab nations

Cross-country teams are allowed and strongly encouraged. A team spanning several countries and disciplines is one of the biggest predictors of a strong, well-rounded solution.



- Who is eligible

Who should apply?

We're looking for multidisciplinary teams that pair data science with environmental domain knowledge. You do not need a space background, curiosity and complementary skills matter more.



Researchers

Environmental scientists and Earth-observation specialists



Data Scientists

ML and AI engineers working with geospatial and hyperspectral data



Engineers

Software, systems and remote-sensing engineers



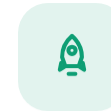
Students

University students committed to a multi-month program



Environmental Experts

Domain specialists in climate, water, ecosystems and GeoAI



Entrepreneurs

Innovators who want to build deployable space solutions

- Building your team

The team rules, and what a strong team looks like



1 to 5 members per team

Solo applicants are welcome, you can be matched with a team later.



Cross-country teams allowed

Members can come from different Arab nations.



At least one technical member

EO, remote sensing, GIS, AI/ML, data science, software or cloud. Basic Python is expected within the team.



30%+ youth

At least 30% of the team aged 35 or under.

Example team · "Team Nur" complementary skills beat four identical profiles ·



GIS student

Jordan



Data scientist

Egypt



Environmental scientist

UAE



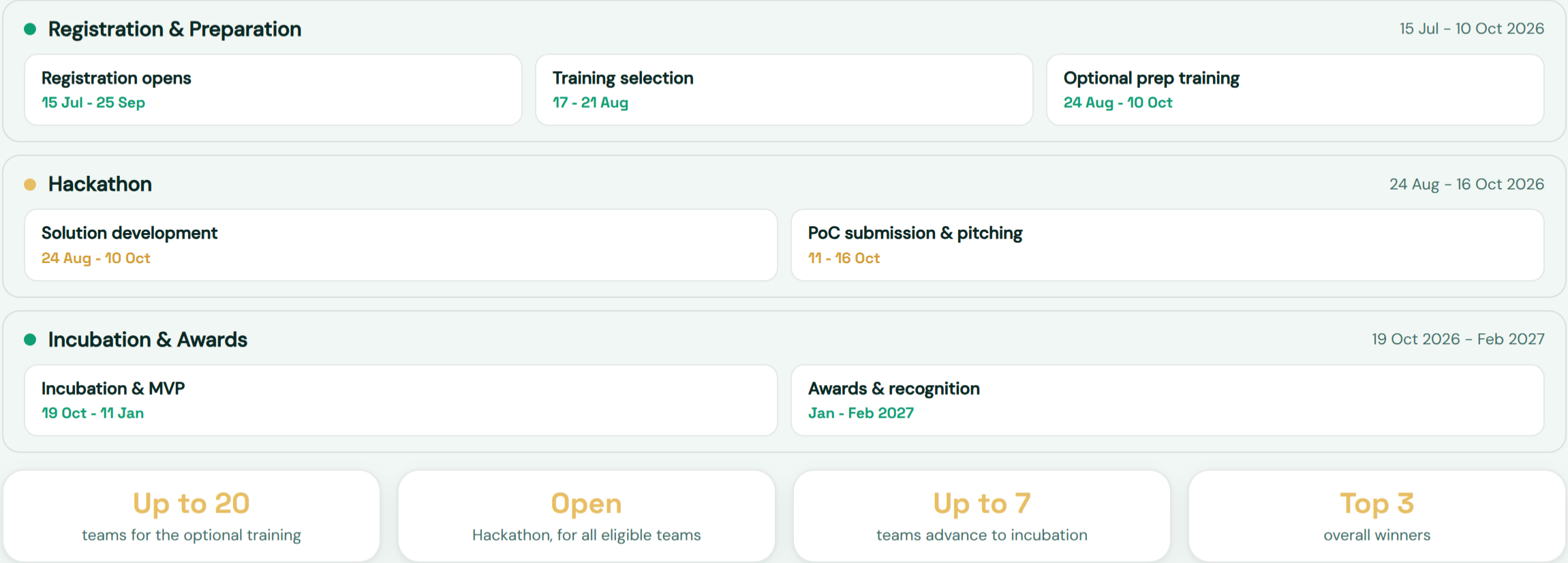
UI designer

Morocco

• The journey

One journey, from an idea to a deployable product

Two things run in parallel: an **optional, competitive preparatory training** for up to 20 teams, and the **Hackathon itself, open to every eligible team**. Training is a head start, not a requirement.



- What's unique

Why this is different from a typical hackathon



A real innovation pathway

Months from concept to a deployable MVP, with incubation for the best teams, not a one-weekend sprint.



Real, multi-source EO data

Live hyperspectral and very-high-resolution imagery (EnMAP, Planet and more), the same data professionals use.



A professional platform

Hands-on access to Space42's GIQ, the UAE's national geospatial platform, for analytics and deployment.



Mentoring at every step

One-to-one guidance from EO, hyperspectral, AI and entrepreneurship experts.



Regional visibility

Exposure to space agencies, industry and policymakers across the Arab world.



Real pathways beyond

Pilots, partnerships, investor conversations and enrolment in the Academy's Hyperspectral Analytics Programme.

- The challenge

What you'll build

Identify a real regional challenge under one of the six official themes, and build a Space Earth Observation analytics solution: a digital **Proof of Concept** that turns satellite data into actionable intelligence, decision-support or operational services.



A technically sound PoC

Pair one or more EO datasets with AI and geospatial analytics. Aim for solutions that are robust, scalable and clearly valuable, socially, environmentally, economically or scientifically.



Use any data that fits

You are not restricted to one sensor. Combine hyperspectral, VHR, multispectral, SAR, thermal or LiDAR as the problem demands.



The hyperspectral incentive

Extra credit goes to solutions that show the unique value of hyperspectral data, especially Arab Satellite 813. An advantage, not a requirement.

The six themes, with worked examples

Pick the one theme that best fits the problem you want to solve. Each maps to the Sustainable Development Goals.



01 / 06

Climate Resilience & Disaster Recovery

Flash floods in wadis, droughts and extreme-heat events. Multi-temporal EO enables rapid hazard mapping, change detection and risk monitoring.

EXAMPLE

Map flood extent within hours of a wadi flash flood to guide emergency response.

SDG 11, 13



02 / 06

Precision Agriculture & Crop Intelligence

Agriculture under severe water scarcity. Hyperspectral and EO data enable crop-stress detection, soil monitoring and precision irrigation.

EXAMPLE

Flag date-palm water stress from spectra weeks before it is visible on the ground.

SDG 2 6 12 13 15



03 / 06

Marine & Water Security

Harmful algal blooms (Red Tides), salinity and protecting desalination intakes. Spectral detection of water-quality indicators.

EXAMPLE

Detect an algal bloom in the Gulf early and alert a desalination operator.

SDG 6, 14

The six themes, with worked examples

Solutions can be technically sound and simple. Judges reward relevance and clarity as much as sophistication.



04 / 06

Blue Carbon & Mangrove Health

Mangrove extent, ecosystem health and coastal resilience. Hyperspectral red-edge analysis tracks vegetation condition and carbon potential.

EXAMPLE

Monitor a mangrove reserve's health and estimate its carbon sequestration.

SDG 13 14 15



05 / 06

Sustainable Urban Planning & Smart Cities

Rapid urban growth, land-use change and the Urban Heat Island effect. EO maps surface materials, expansion and thermal patterns.

EXAMPLE

Map the hottest districts of a growing city to target heat-mitigation planting.

SDG 9, 11



06 / 06

Air Quality & Environmental Intelligence

Air pollution, trace-gas emissions and aerosols. Hyperspectral and EO data detect pollutants and greenhouse gases.

EXAMPLE

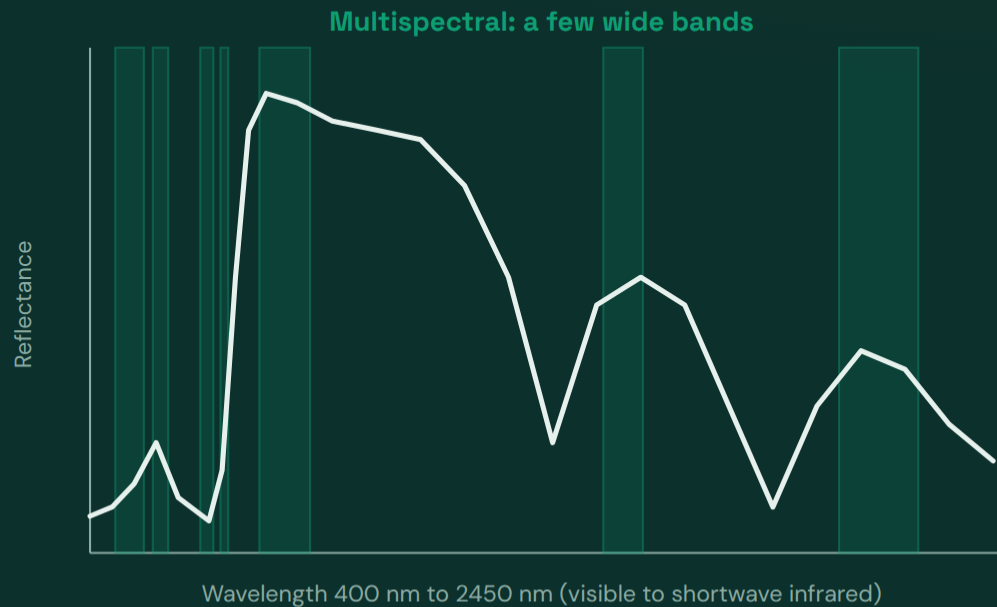
Track NO2 and aerosol hotspots over an industrial zone for public-health alerts.

SDG 3 11 13

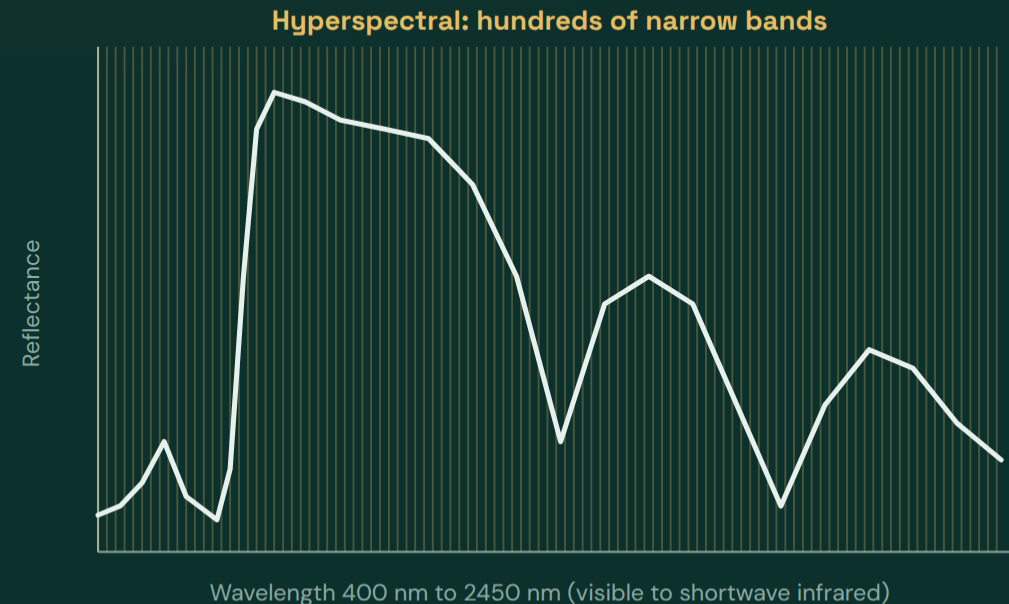
- Data and tools

The idea behind the data: spectral signatures

Every material, healthy or stressed vegetation, clean or turbid water, reflects light differently across wavelengths. **More bands means finer discrimination.** Multispectral samples a few wide bands; hyperspectral samples hundreds of narrow, contiguous ones, giving a near-continuous fingerprint per pixel.



Multispectral (Sentinel-2, Landsat): easy, frequent, and the best starting point for most teams.



Hyperspectral (EnMAP, Satellite 813): reveals material composition, vegetation chemistry, water quality and emissions.

- Data and tools

The imagery you can build with

A multi-source ecosystem, from free global archives to sponsored commercial and hyperspectral scenes. You don't need every dataset, you need the right sensor for the problem.

Sensor	Type	Bands	Resolution	Revisit	Best for
Sentinel-2	Multispectral	13	10–60 m	~5 days	Vegetation, water, land cover, change detection
Landsat 8/9	Multispectral	11	15–100 m	~8–16 days	Historical analysis, temperature, long-term trends
PlanetScope	Multispectral	4–8	~3–4 m	Daily	High-resolution field and urban mapping
MBZ-SAT	Very high-res	–	<1 m	On tasking	Fine spatial detail, local mapping
EnMAP	Hyperspectral	~240	~30 m	~27 days	Ecosystem & scientific spectral analysis
Tanager-1	Hyperspectral	~426	~30 m	Daily	Broad-range spectral analysis, emissions
Satellite 813	Hyperspectral	~205	~20 m	Regional	Water quality, vegetation chemistry, materials

📍 Start with Sentinel-2 or Landsat

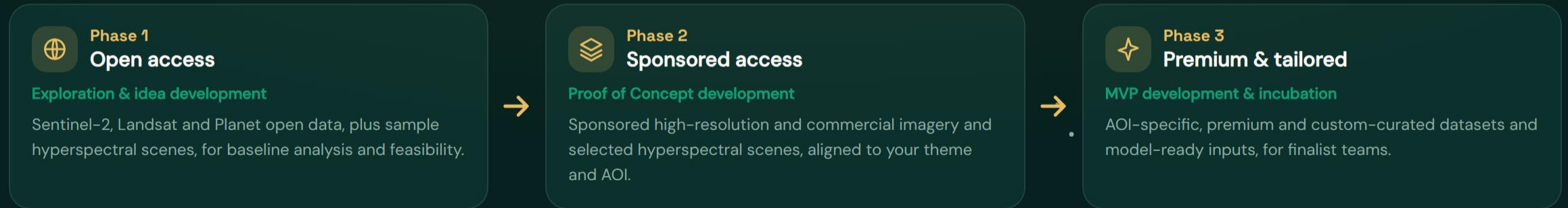
📱 PlanetScope / MBZ-SAT when detail matters

🌟 EnMAP / 813 / Tanager for advanced spectral work

- Data and tools

Data access grows with you

A three-phase model steadily increases what's available as you move from idea, to Proof of Concept, to incubation.



Formats you'll meet:

 GeoTIFF / COG

 HDF5 (hyperspectral)

 STAC metadata

- Data and tools

Your primary workspace: GIQ

GIQ is the UAE's national geospatial platform, operated by Space42, and the primary environment for the program: data access, analytics, AI model development and deployment, all in the cloud. You don't need your own supercomputer.

Teams are free to bring their own preferred tools too:

</> Python & Google Colab

QGIS

</> GitHub

★ AI / ML frameworks

Teams / Slack / Discord

☁ Cloud-native access: work on a small AOI first, and only download what you need.



PLACEHOLDER: GIQ PLATFORM SCREENSHOT

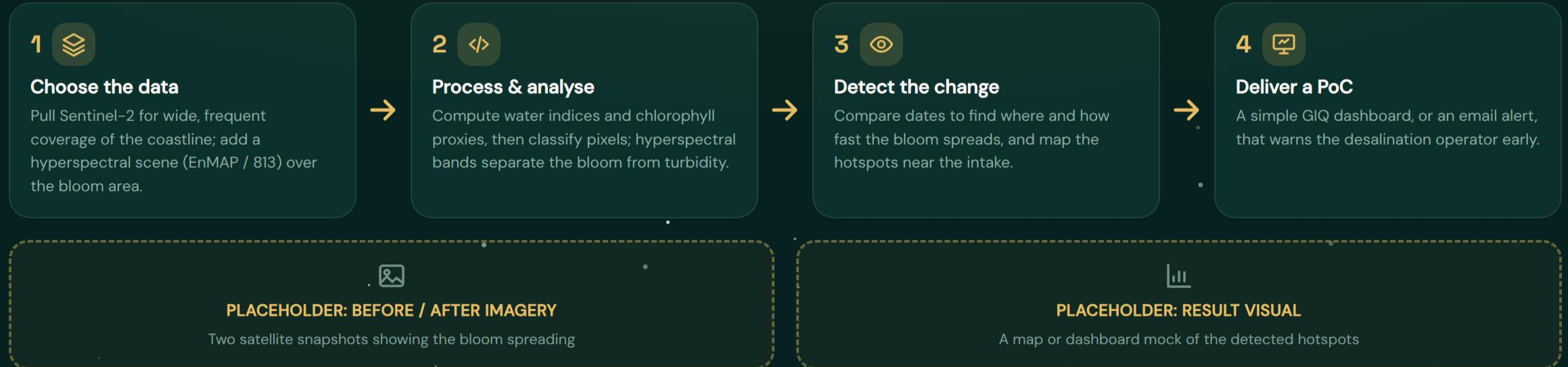
A clean capture of the GIQ workspace or a map/analytics view

GIQ **SPACE42** operated by Space42

- Worked example

From pixels to a Proof of Concept

Theme: **Marine & Water Security**. Goal: detect a harmful algal bloom in the Gulf early enough to protect a desalination intake. Here is the whole path, end to end.



- Criteria for the PoC

What a Proof of Concept is, and what you submit

A PoC is a working, demonstrable slice that shows your idea is technically feasible with the data, not a finished product. Keep the core tight and make it work.



At registration (by 25 Sep)

- ✓ Team details (or solo registration)
- ✓ Country of representation
- ✓ Your chosen challenge theme
- ✓ A short challenge idea or concept
- ✓ Motivation and expected impact
- ✓ (Optional) opt in for the preparatory training

A complete solution isn't required, just a clear, compelling idea.



To be eligible for PoC judging

- ✓ Complete registration by 25 September
- ✓ Receive eligibility confirmation
- ✓ Register all team members
- ✓ Address one official theme
- ✓ Submit your project in English
- ✓ Opt in for judging

PoC deadline: 11 October 2026, 11:59 PM (local time).

- Criteria for the PoC

How your Proof of Concept is scored

Judges assess against these criteria; scoring is individual, with a tie-break on the technical score.



Problem definition

Clarity and real-world relevance of the problem.



Technical soundness

Quality, robustness and scalability of the approach.



Use of hyperspectral / EO data

Meaningful use of the data, with bonus credit for hyperspectral and 813.



Product & delivery model

Insights, decision-support or APIs, and how it would deploy.



Innovation

Novelty relative to existing approaches.



Impact & strategic alignment

Societal, environmental or economic value and SDG fit.



Business viability

A short plan showing economic potential.



Teams that communicate well stand out. A promising solution falls short if its value isn't clear, invest in how you present, not just how you build.

- Registration process

How to apply, in four steps

1



Go to the official website

Open the Arab Youth Space Hackathon platform and create an account.

2



Register your team, or yourself

Teams of 1 to 5. Solo applicants are welcome and can be matched later.

3



Pick a theme & describe your idea

Choose one of the six themes and submit a short concept, motivation and expected impact.

4



(Optional) apply for training

Want a head start? Opt in for the preparatory training, by 16 August 2026.

Mark these dates

16 Aug 2026

Optional training deadline

25 Sep 2026

Hackathon registration deadline

11 Oct 2026

PoC submission deadline



The entire journey is fully online. No travel required at any core stage.

هَاكَاثُون
الشَّبَابِ الْعَرَبِيِّ
THE ARAB YOUTH HACKATHON



Questions?

15 minutes of live Q&A · Driving impact through innovation

Collaborate, learn, work with real space data, and build a solution with real impact for the region. We can't wait to see what you create.

📍 Register by 25 September 2026

✦ Optional training by 16 August 2026



PLACEHOLDER:
REGISTRATION
QR

Link to the
official platform

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